

Bounds on the Interchannel Interference of OFDM in Time-Varying Impairments

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Abstract—In this letter, tight and universal bounds have been derived for the interchannel interference (ICI) of an orthogonal frequency-division multiplexing (OFDM) signal resulting from Doppler spread. The universal bound depends only on the product of the maximum Doppler frequency and the OFDM symbol duration. The tight bound also depends on the variance of the Doppler spectrum. Compared with the exact ICI expressions derived by other researchers, these bounds are easier to evaluate and can provide useful insight.

I. INTRODUCTION

IN ORTHOGONAL frequency-division multiplexing (OFDM) [1], the transmission bandwidth is divided into many narrow subchannels, which are transmitted in parallel. Therefore, the symbol duration is increased and the intersymbol interference (ISI) caused by a time-dispersive fading environment is mitigated. However, with the longer symbol duration, the interchannel interference (ICI) caused by Doppler spread in mobile wireless channels will be increased. As indicated in [2], the ICI degrades the performance of OFDM systems. Several researchers (for example, see [2]–[5] and the references therein) have studied the effect of Doppler spread on an OFDM signal. In particular, Russell and Stüber [2] and Robertson and Kaiser [3] have obtained exact expressions for the ICI of an OFDM signal resulting from Doppler spread. Here, we derive bounds on the ICI which are easier to calculate and which can provide useful insight. Both tight and universal bounds are obtained. The universal bound depends only on the maximum Doppler frequency (f_d) and the symbol duration (T_s). The tight bound is also dependent on the variance of the Doppler spectrum.

II. OFDM OVER TIME-VARYING CHANNELS

Consider an OFDM signal represented in the time domain as

$$x(t) = \sum_k s[k] e^{j2\pi f_k t}, \quad 0 \leq t \leq T_s \quad (2.1)$$

where $f_k = f_o + k\Delta f$ is the frequency of the k th subchannel and $\Delta f = 1/T_s$. The signal transmitted over the k th subchannel $s[k]$ is assumed to be independent for different subchannels. The received signal, after passing through a time-varying channel with a impulse response $h(t, \tau)$, is

$$\tilde{x}(t) = \int h(t, \tau) x(t - \tau) d\tau$$

where additive channel noise and cochannel interference have been ignored.

For flat fading conditions, the channel-impulse response can be represented as

$$h(t, \tau) = \gamma(t)\delta(\tau)$$

and the received signal becomes

$$\tilde{x}(t) = \gamma(t)x(t) \quad (2.2)$$

where $\gamma(t)$ is a wide-sense stationary stochastic process with zero mean and unit variance. For the classical Doppler spectrum [6], the spectral density of $\gamma(t)$ is

$$P_J(f) = \begin{cases} \frac{1}{\pi f_d} \frac{1}{\sqrt{1 - \left(\frac{f}{f_d}\right)^2}}, & \text{if } |f| < f_d \\ 0, & \text{otherwise} \end{cases}$$

where f_d is the maximum Doppler frequency. Two extreme cases of the Doppler spectrum are the uniform and the two-path models, which have been studied in [3]. For these two models

$$P_u(f) = \begin{cases} \frac{1}{2f_d}, & \text{if } |f| < f_d \\ 0, & \text{otherwise} \end{cases}$$

and

$$P_t(f) = \frac{1}{2}[\delta(f + f_d) + \delta(f - f_d)]$$

respectively. The correlation function of $\gamma(t)$, defined as $r(\tau) = E\{\gamma(t + \tau)\gamma^*(t)\}$, is easily obtained as

$$r(\tau) = \mathcal{F}^{-1}\{P(f)\}.$$

The correlation functions for the three models given above are

$$r_J(\tau) = J_0(2\pi f_d \tau) \quad (\text{classical})$$

$$r_u(\tau) = \text{sinc}(f_d \tau) \quad (\text{uniform})$$

and

$$r_t(\tau) = \cos(2\pi f_d \tau) \quad (\text{two-path})$$

where $J_0(x)$ is the zeroth-order Bessel function of the first kind and $\text{sinc}(x) \triangleq \sin(\pi x)/\pi x$. It should be noted that the two-path model corresponds to an OFDM system with a fixed frequency offset of f_d Hz.

For time-dispersive fading channels

$$h(t, \tau) = \sum_i \gamma_i(t)\delta(\tau - \tau_i)$$

and

$$\tilde{x}(t) = \sum_i \gamma_i(t)x(t - \tau_i) \quad (2.3)$$

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where τ_i is the delay of the i th path and $\gamma_i(t)$ is the corresponding complex amplitude. In this letter, we will assume that the complex stochastic processes $\gamma_i(t)$'s are independent for different i 's, but have the same statistics. In addition, we also assume that the energy is normalized so that

$$\sum_i E\{|\gamma_i(t)|^2\} = 1.$$

III. ICI ANALYSIS

The demodulated signal can be obtained from the received signal $\tilde{x}(t)$ as

$$\tilde{s}[m] \triangleq \frac{1}{T_s} \int_0^{T_s} \tilde{x}(t) e^{-j2\pi f_m t} dt. \quad (3.1)$$

For simplicity, an integration is used here instead of the *discrete Fourier transform* (DFT). Because of this, the results obtained in this section apply to the case of an infinite number of carriers. However, compared with the case with a finite number of carriers in [3], the difference is negligible.

A. Exact Calculation

For simplicity, we first derive the ICI bounds for flat fading channels, and then extend the results to frequency-selective environments. From (2.1), (2.2), and (3.1), the demodulated signal becomes

$$\begin{aligned} \tilde{s}[m] &= \frac{1}{T_s} \int_0^{T_s} \gamma(t) \sum_k s[k] e^{j2\pi f_k t} e^{-j2\pi f_m t} dt \\ &= \sum_k \left\{ \frac{1}{T_s} \int_0^{T_s} \gamma(t) e^{-j2\pi(f_m - f_k)t} dt \right\} s[k] \\ &= a_0 s[m] + \underbrace{\sum_{k \neq m} a_{m-k} s[k]}_{\text{ICI}} \end{aligned} \quad (3.2)$$

where a_l is defined as

$$a_l \triangleq \frac{1}{T_s} \int_0^{T_s} \gamma(t) e^{-j2\pi l \Delta f t} dt. \quad (3.3)$$

a_0 represents the attenuation and phase shift of the desired signal and the a_l 's, for $l \neq 0$, represent the channel gains of the interfering signals. With any time variation in the channel, $a_l \neq 0$, and ICI results.

The ICI power is defined as

$$P_{\text{ICI}} \triangleq E \left| \sum_{l \neq 0} a_l s[m-l] \right|^2$$

and, as shown in the Appendix, for OFDM with an infinite number of subcarriers, P_{ICI} can be expressed in terms of the correlation $r(\tau)$ as

$$P_{\text{ICI}} = \int_{-1}^1 (1 - |x|)(1 - r(T_s x)) dx \quad (3.4)$$

or in terms of the Doppler spectral density as

$$P_{\text{ICI}} = 1 - \int_{-f_d}^{f_d} P(f) \text{sinc}^2(f T_s) df. \quad (3.5)$$

Even though (3.4) and (3.5) are for OFDM with a finite number of subcarriers, compared with [2] we can see with only negligible error they can be used for OFDM with finite number of subcarriers. Therefore, (3.4) and (3.5) can be used as a very good approximation of the ICI power for OFDM with finite number of subcarriers.

Once the time-domain correlation of the time-varying channel $r(\tau)$ is known, the ICI power can be calculated using (3.4). For the classical model

$$P_{\text{ICI}} = 1 - \int_{-1}^1 (1 - |x|) J_0(2\pi f_d T_s x) dx \quad (3.6)$$

which was previously derived by Russell and Stüber [2]. For the uniform and two-path models

$$P_{\text{ICI}} = 1 - \frac{1 - \cos(2\pi f_d T_s) - 2\pi f_d T_s \text{Si}(2f_d T_s)}{2(\pi f_d T_s)^2}, \quad (3.7)$$

and

$$P_{\text{ICI}} = 1 - \text{sinc}^2(f_d T_s) \quad (3.8)$$

respectively, where

$$\text{Si}(x) = \pi \int_0^x \frac{\sin(\pi u)}{\pi u} du = \pi \int_0^x \text{sinc}(u) du.$$

The expressions in (3.7) and (3.8) were previously derived by Robertson and Kaiser [3].

B. Bounds

Using the expressions derived above, the ICI power can be exactly calculated. However, the exact expressions are complicated and do not easily provide much insight. Furthermore, in many instances, the exact time-domain correlation or power spectrum is not available. Here, we derive tight upper and lower bounds and a universal bound on the ICI power. These bounds are less complicated and the insight is more readily obtained.

First, express $1 - r(T_s x)$ in (3.4) in terms of the power spectrum $P(f)$ as

$$\begin{aligned} 1 - r(T_s x) &= \int_{-f_d}^{f_d} P(f) (1 - e^{j2\pi f T_s x}) df \\ &= 2 \int_0^{f_d} P(f) (1 - \cos(2\pi f T_s x)) df. \end{aligned}$$

It can be easily shown that

$$\frac{1}{2} \theta^2 - \frac{1}{24} \theta^4 \leq 1 - \cos \theta \leq \frac{1}{2} \theta^2.$$

Therefore

$$1 - r(T_s x) \geq \frac{\alpha_1}{2} (2\pi f_d T_s)^2 x^2 - \frac{\alpha_2}{24} (2\pi f_d T_s)^4 x^4$$

and

$$1 - r(T_s x) \leq \frac{\alpha_1}{2} (2\pi f_d T_s)^2 x^2$$

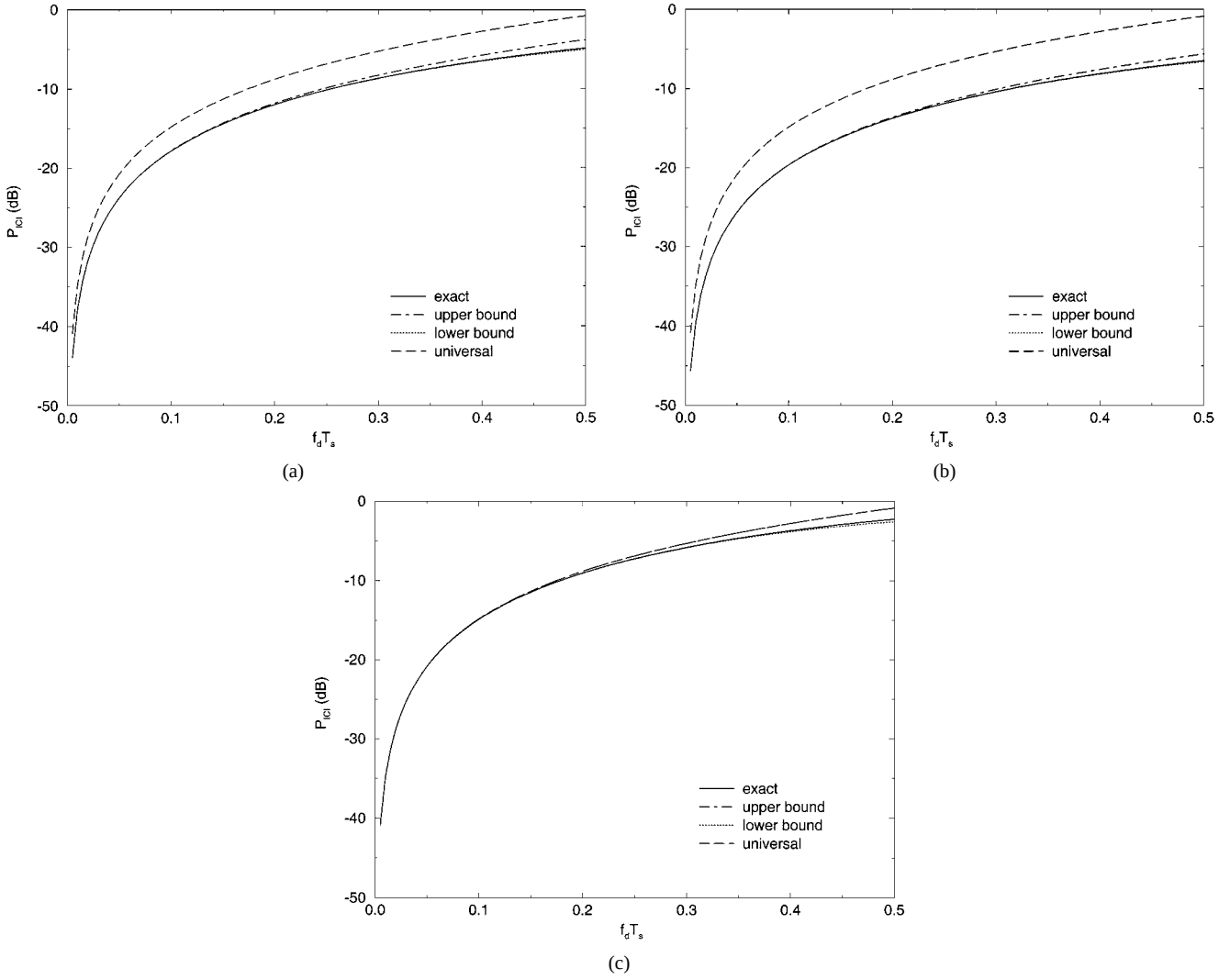


Fig. 1. Comparison of exact ICI power, tight and universal bounds for (a) the classical model, (b) the uniform model, and (c) the two-path model.

TABLE I
 α_1 'S AND α_2 'S FOR DIFFERENT TIME-VARYING MODELS

model	α_1	α_2
Classical	1/2	3/8
Uniform	1/3	1/5
Two-path	1	1

where α_i 's, for $i = 1, 2$, are defined as

$$\alpha_k \triangleq \frac{1}{f_d^{2k}} \int_{-f_d}^{f_d} f^{2k} P(f) df = \frac{2}{f_d^{2k}} \int_0^{f_d} f^{2k} P(f) df.$$

The constants α_1 and α_2 are easy to calculate and are given in Table I for the models of interest in this letter. Substituting the above inequality into (3.4), we obtain bounds on the ICI power

$$P_{ICI} \geq \frac{\alpha_1}{12} (2\pi f_d T_s)^2 - \frac{\alpha_2}{360} (2\pi f_d T_s)^4 \quad (3.9)$$

and

$$P_{ICI} \leq \frac{\alpha_1}{12} (2\pi f_d T_s)^2. \quad (3.10)$$

Without knowing the Doppler spectrum, α_1 and α_2 can be also evaluated using other approaches. For example, it has been proved in [7] that

$$E \left\{ \left| \gamma^{(k)}(t) \right|^2 \right\} = (2\pi)^{2k} \int_{-f_d}^{f_d} f^{2k} P(f) df$$

where $\gamma^{(k)}(t) = d^k \gamma(t) / dt^k$. Then, α_k can be evaluated by

$$\alpha_k = \frac{E \left\{ \left| \gamma^{(k)}(t) \right|^2 \right\}}{(2\pi f_d)^{2k}}$$

which is much simpler than obtaining by the Doppler spectrum.

From the definition of α_1 , it is clear that $\alpha_1 \leq 1$. Using this fact, together with (3.10), we can obtain a *universal upper bound* on the ICI power, that is dependent only on $f_d T_s$

$$P_{ICI} \leq \frac{1}{12} (2\pi f_d T_s)^2. \quad (3.11)$$

This universal upper bound can be used in OFDM systems with any Doppler spectra, including OFDM systems with frequency offset. Since α_1 is usually less than 1, the above bound is looser

than the bound in (3.10). However, (3.11) is much easier to calculate since it only depends on $f_d T_s$. For the two-path model, $\alpha_1 = 1$, and the universal bound is also the tight bound.

In the above discussions, we have obtained tight and universal bounds for flat fading channels. For OFDM systems in time-dispersive fading channels, it is usually assumed that a cyclic extension is inserted between OFDM symbols to prevent interblock interference. Under this assumption, we have

$$a_l = \sum_i \frac{e^{j2\pi l \Delta f \tau_i}}{T_s} \int_0^{T_s} \gamma_i(t) e^{-j2\pi l \Delta f t} dt.$$

Since the $\gamma_i(t)$'s for different i 's are independent and have the same statistics, $E\{|a_l|^2\}$ is the same for both flat and dispersive fading channels. Therefore, the exact expressions for P_{ICI} and the various bounds are also applicable to dispersive channels.

IV. RESULTS

To get insight into the accuracy of the bounds, in Fig. 1, we compare the upper and the lower tight bounds and the universal bound with the exact value of the ICI power, P_{ICI} . Note that for the two-path model, Fig. 1(c), the upper tight bound and the universal bound are identical. From Fig. 1, the tight bounds are very close to the exact P_{ICI} . When designing OFDM systems, if symbol duration T_s is chosen so that $f_d T_s$ is very small, then the effect of the Doppler spread is negligible. In the design example in [8], $T_s = 160 \mu\text{s}$, then $P_{\text{ICI}} < 0.168\%$ (-27.7 dB) when $f_d = 200 \text{ Hz}$. Therefore, the P_{ICI} is much less than the noise or cochannel interference level.

APPENDIX

DERIVATION OF EXACT EXPRESSION OF P_{ICI}

The ICI power can be expressed as

$$P_{\text{ICI}} = E \left| \sum_{l \neq 0} a_l s[m-l] \right|^2 = \sum_{l \neq 0} E\{|a_l|^2\} \quad (\text{A.1})$$

where we have used the assumption that $s[m]$'s are independent for different m 's and the definition of a_l in (3.3).

Using (3.3) and [7, eqs. (10)–(50)],

$$E\{|a_l|^2\} = \int_{-1}^1 r(T_s x) (1 - |x|) e^{-j2\pi l x} dx. \quad (\text{A.2})$$

Therefore, the total ICI power due to Doppler spread is

$$\begin{aligned} P_{\text{ICI}} &= \sum_{l \neq 0} \int_{-1}^1 r(T_s x) (1 - |x|) e^{-j2\pi l x} dx \\ &= \int_{-1}^1 r(T_s x) (1 - |x|) \left(\sum_{l \neq 0} e^{-j2\pi l x} \right) dx. \end{aligned}$$

From [9, eqs. (3.2) and (3.5)]

$$\sum_{l=-\infty}^{+\infty} e^{-j2\pi l x} = \sum_{m=-\infty}^{+\infty} \delta(x - m).$$

Consequently

$$\begin{aligned} P_{\text{ICI}} &= \int_{-1}^1 r(T_s x) (1 - |x|) \left(\sum_m \delta(x - m) - 1 \right) dx \\ &= \int_{-1}^1 (1 - |x|) (1 - r(T_s x)) dx \quad (\text{A.3}) \end{aligned}$$

or

$$\begin{aligned} P_{\text{ICI}} &= \int_{-1}^1 (1 - |x|) \left(1 - 2 \int_0^{f_d} P(f) \cos(2\pi f T_s x) df \right) dx \\ &= \int_{-1}^1 (1 - |x|) dx \\ &\quad - 2 \int_0^{f_d} \int_{-1}^1 (1 - |x|) P(f) \cos(2\pi f T_s x) dx df \\ &= 1 - 2 \int_0^{f_d} P(f) \text{sinc}^2(f T_s) df. \end{aligned}$$

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